

Mini Project - Pedestrian Crossing Analysis

Raw Data Details

- 1) pedestrian-counting-system-sensor-locations.xlsx: sensor master data (LocationID, SensorName, LocationType, Status, Latitude, Longitude, etc.).
- 2) pedestrian-counting-system-trimmed-2.xlsx: pedestrian counts (ID, LocationID, SensingDate, HourDay, Direction1, Direction2, TotalofDirections, etc.).

Data Pre-processing using Excel

1) Removing duplicates

ID	LocationID	Sensor	Description	Sensing Date	Location
1	3	Swab20	T	2009-03-25	Outdoor A North South
2	9	Princes Bridge	Foot	2009-03-25	Outdoor A North South
3	9	Southern Cross Station	Car	2009-03-25	Outdoor A East West
4	14	Sandridge Bridge	SanBri	2009-03-24	Outdoor A North South
5	18	Coltine Place (North)	ColLT	2009-03-20	Outdoor A North South
6	19	Christians-Sanderton St (North)	CRSD10	2013-09-02	Outdoor A East West
7	20	Christians-La Bourke St (South)	CRSD10	2013-09-06	Outdoor A East West
8	21	Flinders-La Sanderton St (West)	FLSD10	2013-08-29	Outdoor A North South
9	43	Monash Rd Sanderton St (West)	UMH	2013-04-15	Outdoor A North South
10	44	Tin Alley Sanderton St (West)	UMH	2013-04-15	Outdoor A North South
11	46	Palham St (South)	PalAT	2013-07-10	Outdoor A East West
12	47	Malbourne Central-Elizabeth St (East)	ELSD10	2013-08-24	Outdoor A North South
13	48	QVM Queen St (East)	QVMQ	2013-08-02	Outdoor A East West
14	50	Faraday St Upton St (North)	Lyg100	2013-11-30	Outdoor A North South
15	51	QVM Franklin St (North)	FrankLT	2013-11-30	Outdoor A East West
16	52	Elizabeth St Lonsdale St (South)	ELSD10	2013-08-02	Outdoor A East West
17	53	Coltine Street (North)	ColST4	2010-09-23	Outdoor A East West
18	54	Lincoln Sanderton St (West)	Swab20	2010-08-26	Outdoor A North South
19	56	Lonsdale St - Elizabeth St (North)	LondST4	2010-07-25	Outdoor A East West
20	68	Flinders Ln - Engraves St (North)	FLDngLT	2010-06-03	Outdoor A East West
21	69	Flinders Ln - Engraves St (Crossing)	FLDngLT	2010-06-03	Outdoor A North South
22	72	Flinders St ACMI	ACMI	2010-11-30	Outdoor A East West
23	81	Boyd Community - Front door	BoCoF	2012-03-03	Indoor A East West
24	82	112 Elizabeth Street	ELSD10	2013-11-06	Indoor A East West
25	85	Macaulay Rd (North)	488Mac	2012-12-21	Outdoor A East West
26	86	Queensberry St - Ernst St (South)	ST4QuB	2012-12-21	Outdoor A East West
27	89	City Library	CityLT	2013-12-11	Indoor A East West
28	90	Boyd Community Hub - Library	BoCoL	2010-08-11	Indoor A East West
29	93	East Melbourne Library	EastLib	2013-02-02	Indoor A East West
30	96	Flinders Garden Visitor Centre Internal	FlindersLT	2013-02-02	Indoor A East West
31	105	Kathleen Syme Library Lib	KathLib	2010-02-02	Indoor A East West
32	108	William St - Little Lonsdale St (West)	251WLT	2012-03-09	Outdoor A North South
33	116	Flinders Garden Visitor Centre Cafe Viewside	FLDngLT	2013-02-02	Indoor A East West
34	117	114 Flinders Street Car Park Footpath	FLDngLT	2012-12-06	Outdoor A East West
35	123	Brimarum Main East - Batman Ave Bridge Entry	BrimBridge	2012-06-02	Outdoor A East West
36	132	1-Hub Corner of King Street and Lonsdale Street (137-205 King)	King205	2012-09-25	Outdoor A North South
37	133	1-Hub Southern Cross Station - Lonsdale Street Entrance - South	Spw220	2012-09-25	Outdoor A North South
38	136	CONP Point 1120 - Towards the City, Near Federation Square Bridge North	FLDngLT	2012-11-03	Outdoor A East West

2) Handling missing values

This is done by filtering for blanks in any of the columns, if any and deleting the rows to exclude from further analysis

ID	LocationID	Sensing Date	HourDay	Direction1	Direction2	Total of Directions	Sensor	Location
1	6.92E+11	69	2025-04-01	17	14	226	144.96509144	
2	6.92E+11	69	2026-06-01	23	1	226	144.96509144	
3	6.92E+11	69	2025-05-2	17	14	226	144.96509144	
4	8.48E+10	84	2026-01-11	8	27	449	144.96503383	
5	8.42E+11	84	2025-01-01	15	66	449	144.96503383	
6	8.49E+10	84	2024-07-11	9	111	449	144.96503383	
7	8.41E+10	84	2026-03-11	1	4	449	144.96503383	
8	4.15E+10	4	2026-04-01	15	146	888	144.96608078	
9	4.32E+09	4	2026-05-21	3	4	888	144.96608078	
10	7.02E+11	70	2026-01-21	16	11	884	144.94846228	
11	7.06E+10	70	2024-10-01	6	1	884	144.94846228	
12	6.62E+10	66	2025-06-01	2	2	346	144.96444294	
13	6.66E+10	66	2024-10-01	0	2	346	144.96444294	
14	3.13E+10	3	2024-10-11	13	92	324	144.96429485	
15	3.13E+10	3	2026-03-2	13	126	324	144.96429485	
16	7.52E+11	75	2024-07-01	23	3	276	144.97467661	
17	7.58E+10	75	2025-05-21	8	2	276	144.97467661	
18	7.59E+10	75	2025-12-2	9	2	276	144.97467661	
19	4.25E+10	42	2025-03-11	5	2	966	144.96386412	
20	4.25E+10	42	2025-09-21	5	2	966	144.96386412	
21	7.79E+10	77	2024-09-1	19	49	336	144.94433026	
22	7.72E+11	77	2025-03-11	17	7	336	144.94433026	
23	4.29E+10	42	2025-01-01	2	3	17	144.96672865	
24	1.42E+11	14	2025-12-01	21	100	342	144.96291897	
25	1.47E+10	14	2025-02-2	7	159	128	323	SanBri
26	1.41E+11	14	2026-03-26	13	221	114	335	SanBri
27	1.42E+11	14	2026-06-01	22	37	5	42	SanBri
28	5.9E+10	59	2025-11-11	0	100	46	146	RMIT
29	5.92E+10	59	2025-03-2	2	50	16	66	RMIT
30	5.91E+11	59	2025-10-01	12	1097	1096	2193	RMIT
31	5.93E+10	59	2024-08-01	3	22	11	33	RMIT
32	7.12E+11	71	2025-11-30	23	2	1	3	WestWP
33	4.02E+11	40	2025-09-01	19	49	44	93	Spr201
34	4.02E+11	40	2025-12-2	20	255	51	306	Spr201
35	4.02E+11	40	2026-05-1	18	79	101	180	Spr201
36	4.02E+10	40	2025-01-01	2	3	1	4	Spr201
37	3.1E+10	31	2025-03-21	0	5	7	12	Lyg161
38	5.41E+11	54	2025-10-31	10	144	89	233	Swab20

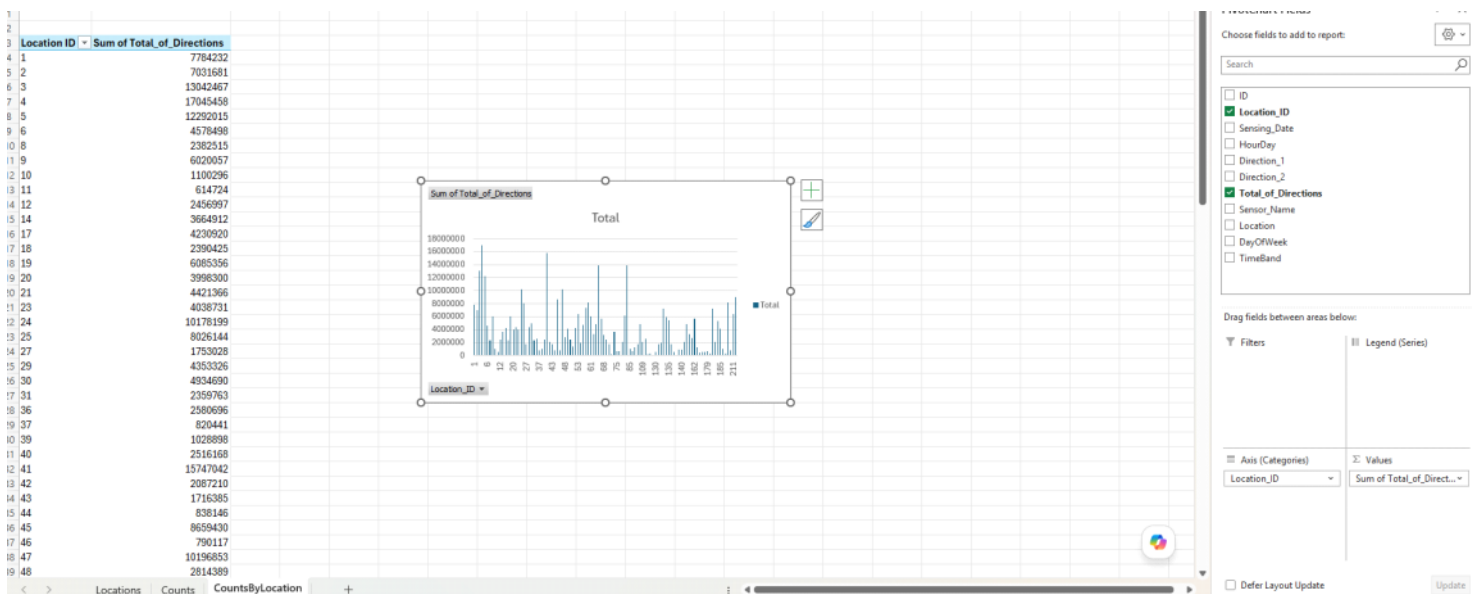
3) Standardize formats

=IF([@HourDay]<12,"Morning",IF([@HourDay]<18,"Afternoon","Evening"))										
ID	Location_ID	Sensing_Date	HourDay	Direction_1	Direction_2	Total_of_Directions	Sensor_Name	Location	DayOfWeek	TimeBand
6.9172E+11	69	2025-04-02	17	147	231	378	FLDegC_T	-37.81687226, 144.96559144	Wed	Afternoon
6.9232E+11	69	2026-06-06	23	16	32	48	FLDegC_T	-37.81687226, 144.96559144	Sat	Evening
6.9172E+11	69	2025-05-23	17	145	214	359	FLDegC_T	-37.81687226, 144.96559144	Fri	Afternoon
8.482E+10	84	2026-01-18	8	276	287	563	EiFl_T	-37.81798049, 144.96503383	Sun	Morning
8.4152E+11	84	2025-01-05	15	685	865	1550	EiFl_T	-37.81798049, 144.96503383	Sun	Afternoon
8.492E+10	84	2024-07-18	9	1113	362	1475	EiFl_T	-37.81798049, 144.96503383	Thu	Morning
8.412E+10	84	2026-03-10	1	29	31	60	EiFl_T	-37.81798049, 144.96503383	Tue	Morning
4.152E+10	4	2026-04-01	15	1495	1268	2763	Swa123_T	-37.81487988, 144.9660878	Wed	Afternoon
4320260529	4	2026-05-29	3	41	29	70	Swa123_T	-37.81487988, 144.9660878	Fri	Morning
7.0162E+11	70	2026-01-22	16	157	117	274	Errol20_T	-37.80456984, 144.94946228	Thu	Afternoon
7.062E+10	70	2024-10-07	6	11	9	20	Errol20_T	-37.80456984, 144.94946228	Mon	Morning
6.622E+10	66	2025-06-02	2	25	8	33	QVN_T	-37.81057846, 144.96444294	Mon	Morning
6.802E+10	66	2024-10-04	0	72	81	153	QVN_T	-37.81057846, 144.96444294	Fri	Morning
3.132E+10	3	2024-10-10	13	926	962	1888	Swa295_T	-37.81101524, 144.96429485	Thu	Afternoon
3.132E+10	3	2026-03-21	13	1284	1636	2920	Swa295_T	-37.81101524, 144.96429485	Sat	Afternoon
7.5232E+11	75	2024-07-06	23	5	15	20	SprFl_T	-37.81515276, 144.97467661	Sat	Evening
7.582E+10	75	2025-05-25	8	24	34	58	SprFl_T	-37.81515276, 144.97467661	Sun	Morning
7.592E+10	75	2025-12-23	9	27	26	53	SprFl_T	-37.81515276, 144.97467661	Tue	Morning
4.252E+10	42	2025-03-19	5	6	6	9	UM1_T	-37.80008566, 144.96386412	Wed	Morning
4.252E+10	42	2025-09-22	5	3	0	3	UM1_T	-37.80008566, 144.96386412	Mon	Morning
7.782E+10	77	2024-09-11	8	204	215	419	HarEsP_T	-37.81441438, 144.94433026	Wed	Morning
7.7172E+11	77	2025-03-10	17	79	57	136	HarEsP_T	-37.81441438, 144.94433026	Mon	Afternoon
2.932E+10	29	2025-09-06	3	2	0	2	AG_T	-37.8199817, 144.96872865	Sat	Morning
1.4212E+11	14	2025-12-05	21	1009	200	1209	SanBri_T	-37.82011242, 144.96291897	Fri	Evening
1.472E+10	14	2025-02-21	7	195	128	323	SanBri_T	-37.82011242, 144.96291897	Fri	Morning
1.4132E+11	14	2026-03-20	13	221	114	335	SanBri_T	-37.82011242, 144.96291897	Fri	Afternoon
1.4222E+11	14	2026-06-03	22	37	5	42	SanBri_T	-37.82011242, 144.96291897	Wed	Evening
5.902E+10	59	2025-11-13	0	100	46	146	RMIT_T	-37.80825648, 144.96304859	Thu	Morning
5.922E+10	59	2025-03-23	2	50	16	66	RMIT_T	-37.80825648, 144.96304859	Sun	Morning
5.9122E+11	59	2025-10-03	12	1097	1096	2193	RMIT_T	-37.80825648, 144.96304859	Fri	Afternoon
5.932E+10	59	2024-08-05	3	22	11	33	RMIT_T	-37.80825648, 144.96304859	Mon	Morning

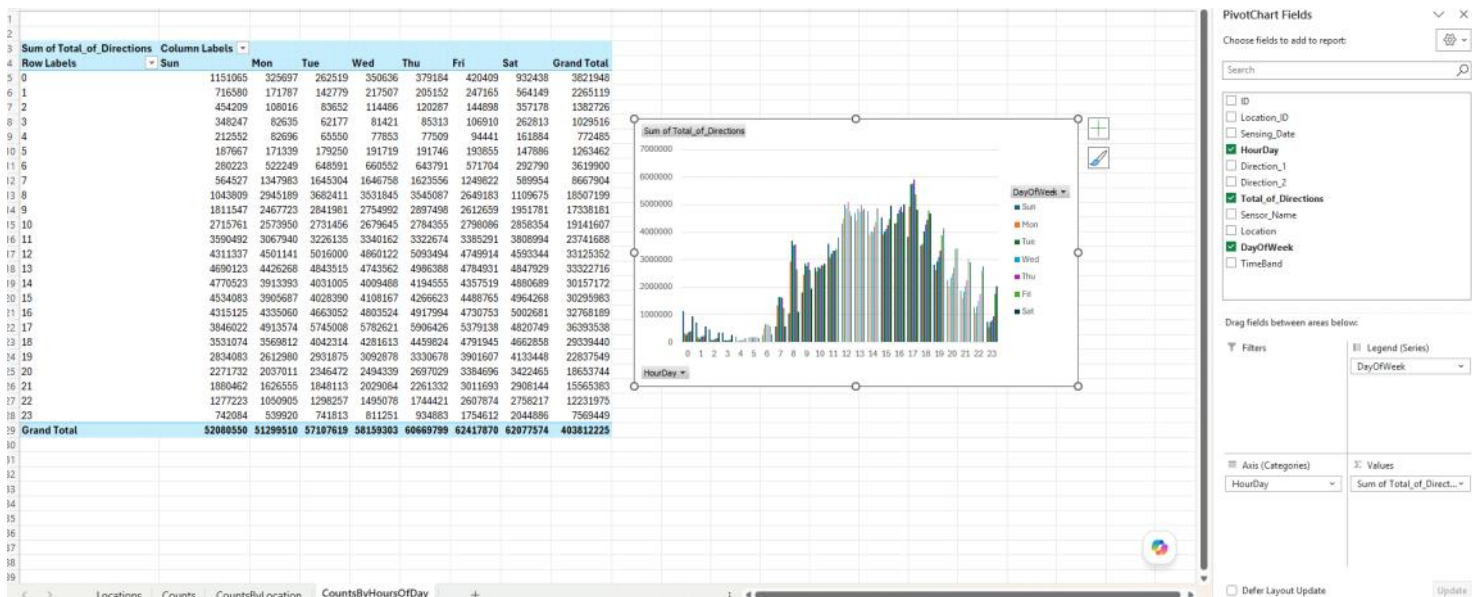
Field: LocationType from Locations sheet

5) Create pivot tables

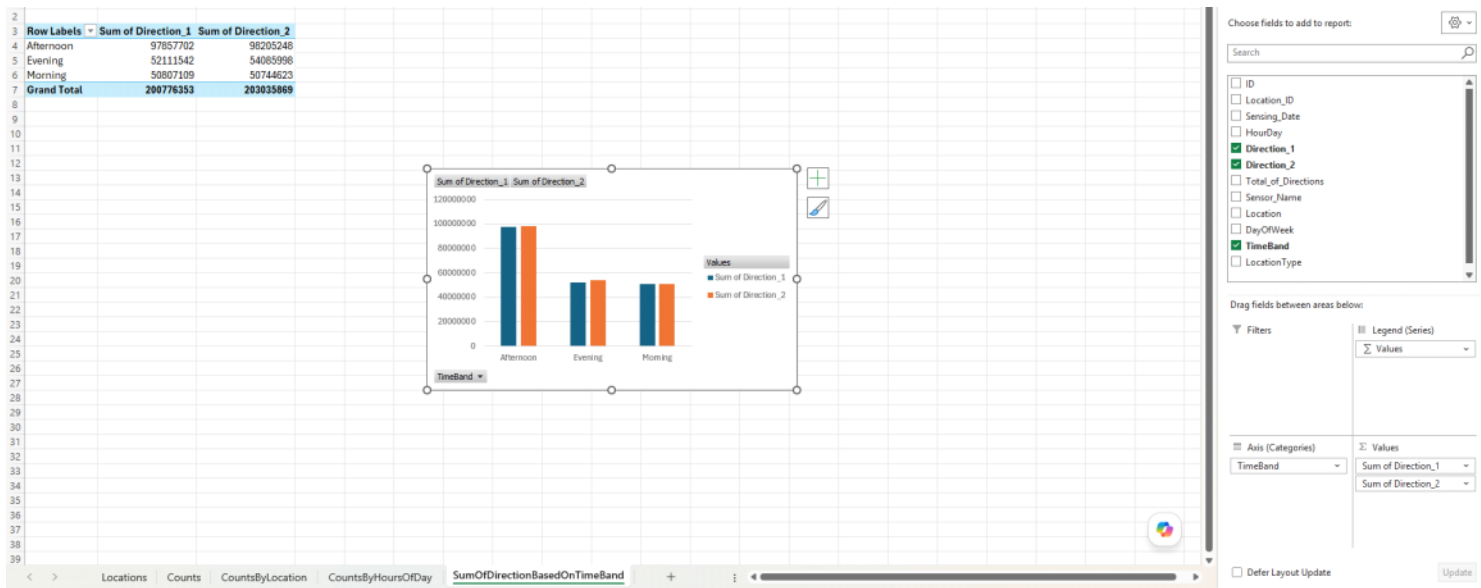
CountsByLocation



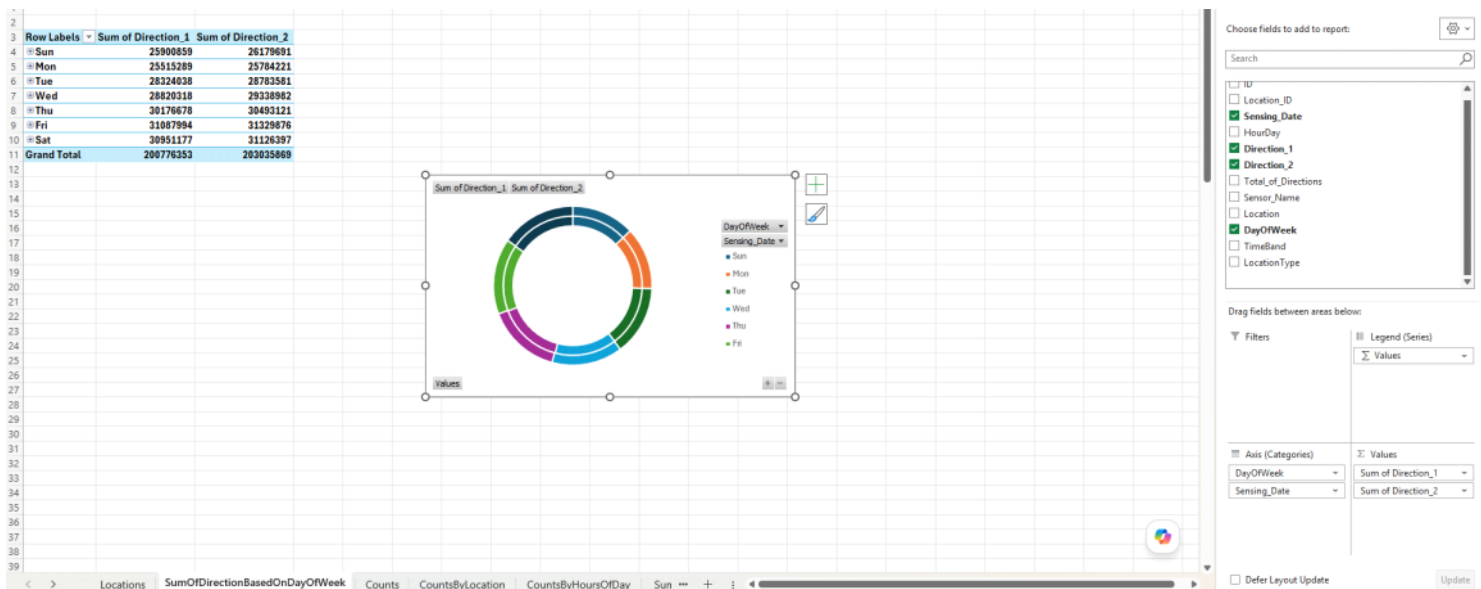
CountsByHoursOfDay



SumOfDirectionBasedOnTimeBand

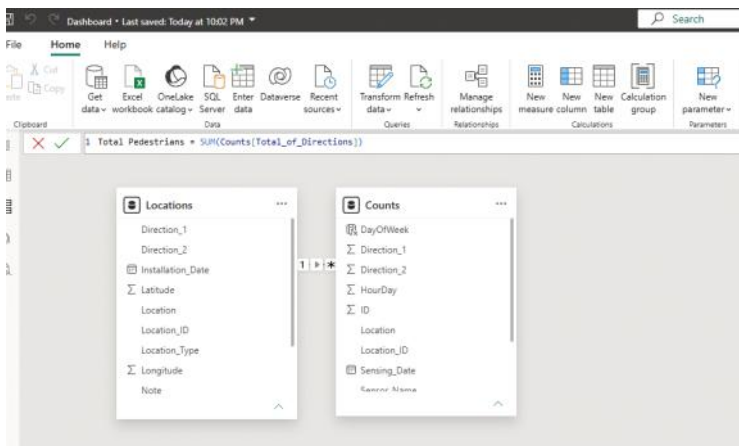


SumOfDirectionBasedOnDayOfWeek



Dashboard using PowerBI

1) Import raw data from excel and ensure LocationID column relationship is established



2) Add calculated and measured column

1 DayOfWeek = FORMAT(Counts[Sensing_Date], "ddd")

The screenshot shows the first DAX measure in the formula bar: `1 DayOfWeek = FORMAT(Counts[Sensing_Date], "ddd")`. The 'Locations' and 'Counts' tables are visible in the model view. The 'Counts' table has columns: DayOfWeek, Direction_1, Direction_2, HourDay, ID, Location, Location_ID, Sensing_Date, and Sensor Name. The 'Properties' pane on the right shows the 'DayOfWeek' table selected, with its name, description, synonyms, display folder, and hidden status.

1 TimeBand = IF(Counts[HourDay] < 12, "Morning", IF(Counts[HourDay] < 18, "Afternoon", "Evening"))

The screenshot shows the second DAX measure in the formula bar: `1 TimeBand = IF(Counts[HourDay] < 12, "Morning", IF(Counts[HourDay] < 18, "Afternoon", "Evening"))`. The 'Locations' and 'Counts' tables are visible in the model view. The 'Properties' pane on the right shows the 'TimeBand' table selected, with its name, description, synonyms, display folder, and hidden status.

1 Total Pedestrians = SUM(Counts[Total_of_Directions])

The screenshot shows the third DAX measure in the formula bar: `1 Total Pedestrians = SUM(Counts[Total_of_Directions])`. The 'Locations' and 'Counts' tables are visible in the model view. The 'Properties' pane on the right shows the 'Total Pedestrians' table selected, with its name, home table, description, synonyms, display folder, and hidden status.

3) Build Core Visuals

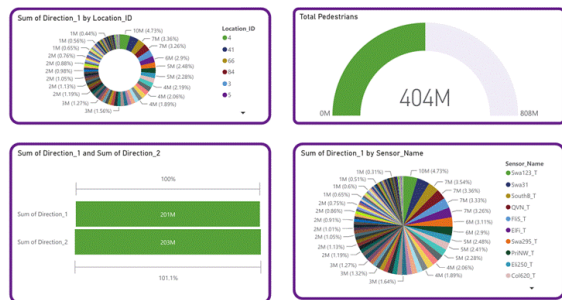
Pedestrian Crossing Trends



Pedestrian Crossing Trends



Pedestrian crossing trends



Pedestrian crossing trends

